# Urban Drone Physics

SKILL.md reference | Remote control, computer-vision navigation, and fibre-tethered multirotors

Consolidated engineering reference. Equations use readable ASCII notation. Illustrative settings are not validated flight limits.


```
---
name: urban-drone-physics
description: Physics-grounded urban multirotor engineering models.
version: 1.0.1
language: en
tags: [drones, urban, dynamics, RF, CV, VIO, fibre-tether]
vehicle_types: [quadrotor, hexarotor, octocopter]
mission_modes: [remote-control, CV-navigation, fibre-tethered]
coordinate_frame: ENU
units: SI unless explicitly stated
assumptions:
  - Body +z is the propulsive thrust direction.
  - Validate aerodynamic and propulsion coefficients experimentally.
  - Urban flow and multipath need site-specific calibration.
  - Distinguish anchored load-bearing tethers from payout fibre.
outputs:
  - thrust-and-energy-margins
  - radio-and-optical-link-budgets
  - visual-navigation-observability
  - tether-loads-and-clearance
  - stopping-and-containment-checks
---
```


## 1. Frames and rigid-body states

### *State and frame transforms*

```
x = [p_W, v_W, q_WB, omega_B]
p_dot = v_W; F_W = R_WB F_B
q_dot = 0.5 * q_WB (quaternion product) [0, omega_B]
```

Use scalar-first Hamilton quaternions and body-to-world rotation consistently. Camera, IMU and tether attachment extrinsics must be calibrated.

## 2. Multirotor dynamics

### *Translation*

```
m v_dot = m g_W + R_WB [0,0,T]^T
          + F_aero + F_tether + F_contact
g_W = [0,0,-g]^T
```

All force vectors on the right are expressed in the world frame.

### *Hover and thrust reserve*

```
T_hover = m g; T_rotor = m g / n
thrust_to_weight = T_max / (m g)
reserve_fraction = (T_max - m g) / (m g)
```

Do not assign universal safety bands to thrust-to-weight ratio. Define reserves from measured vehicle response, power limits and disturbances.

### *Horizontal acceleration in level-altitude flight*

```
tan(theta) = a_h / g
T_required = m sqrt(g^2 + a_h^2)
```

```
a_h,max = sqrt((T_max/m)^2 - g^2)
```

Idealized result: neglects drag, tilt limits, actuator dynamics and tether load.

### *Rotation and rotor allocation*

```
I omega_dot + omega cross (I omega) = tau_total
T_i = k_f omega_i^2; Q_i = k_m omega_i^2
[T,tau_x,tau_y,tau_z]^T = B [omega_1^2,...]^T
```

Build B using actual rotor locations, spin signs and thrust axes. Tether torque is attachment_offset cross tether_force.

## 3. Urban aerodynamics and wind

### *Relative flow and drag*

```
v_rel = v_vehicle - v_wind
F_D = -0.5 rho C_D A |v_rel| v_rel
F_D,axis = -0.5 rho C_D,axis A_axis |v_axis| v_axis
F_D,linear = -D v_rel
```

Choose a calibrated vector or component-wise drag model; do not interchange fitted coefficients. Air density depends on pressure and temperature.

### *Station keeping in crosswind*

```
F_w = 0.5 rho C_D A V_rel^2
theta = atan(F_w/(m g))
T_required = sqrt((m g)^2 + F_w^2)
```

Building corners, rooftop shear, recirculation and facade proximity invalidate a uniform-wind assumption.

### *Gust response*

```
delta_F = 0.5 rho C_D A [(V+delta_V)^2 - V^2]
a_gust = delta_F / m
```

Scalar estimate assumes a gust in the original flow direction. Use vectors for changing wind direction.

### *Rotor induced flow and ground effect*

```
A_disk = pi D_prop^2 / 4
v_i = sqrt(T/(2 rho A_disk))
T_IGE = K_GE(h/D_prop) T_OGE
```

Momentum theory is idealized. Ground and wall effects require platform-specific calibration; no universal clearance threshold is imposed.

## 4. Power and endurance

### *Ideal induced power*

```
A_total = n pi D_prop^2 / 4
P_induced = T^(3/2) / sqrt(2 rho A_total)
P_parasitic = 0.5 rho C_D A V^3
```

Assumes ideal non-interacting actuator disks. Rotor overlap, inflow, profile power and installation losses require corrections.

### *Electrical power and stored energy*

```
P_elec = P_shaft/(eta_motor eta_ESC) + P_avionics + P_payload
E_nominal,Wh = V_nominal C_Ah
E_usable,Wh = E_nominal,Wh DoD eta_battery
t_minutes = 60 E_usable,Wh / P_avg,W
I = P_elec / V_loaded
V_loaded = V_open - I R_internal
```

If induced power rather than shaft power is used, include an explicitly defined aerodynamic correction. Avoid counting efficiency losses twice.

### *Mission energy*

```
E_mission = integral P(t) dt
E_mission + E_return + E_contingency <= E_usable
```

Use consistent joules or watt-hours. Integrating watts over seconds gives joules; divide by 3600 for watt-hours.

## 5. Remote-control radio links

### *Free-space loss and link budget*

```
FSPL_dB = 32.44 + 20 log10(d_km) + 20 log10(f_MHz)
FSPL_dB = 20 log10(4 pi d_m f_Hz / c)
P_r,dBm = P_t + G_t + G_r - L_path - L_system
M_link = P_r - P_sens - M_fade - M_impl
```

Free-space loss is a baseline, not an urban coverage prediction. Include antenna orientation, polarization, shadowing, interference and airframe losses.

### *Urban shadowing and reflections*

```
PL(d) = PL(d0) + 10 n_PL log10(d/d0) + X_sigma
P_r ~ d^(-4)  [asymptotic two-ray regime]
```

Fit path-loss exponent and shadowing distribution to site measurements. Two-ray behavior applies only under suitable geometry and reflection conditions.

### *Fresnel clearance and Doppler*

```
lambda = c/f
r_1 = sqrt(lambda d1 d2/(d1+d2))
planning_clearance >= 0.6 r_1
f_D = (v_radial/c) f_carrier
```

Distances in the Fresnel equation use metres. Clearance is a planning heuristic, not a guarantee against multipath.

### *Latency and stopping*

```
t_delay = t_sensor + t_encode + t_network + t_process + t_actuate
d_latency = v t_delay + 0.5 a t_delay^2
d_stop = v t_delay + v^2/(2 a_brake)
```

Constant-speed delay model in d_stop assumes no acceleration during delay. Otherwise update braking speed to v + a t_delay.

## 6. Computer-vision geometry

### *Projection and intrinsics*

```
u = f_x X_C/Z_C + c_x
v = f_y Y_C/Z_C + c_y
K = [[f_x,0,c_x],[0,f_y,c_y],[0,0,1]]
z_hat = project(T_CW P_W)
```

Projection requires positive camera depth, calibrated extrinsics and consistent timing.

### *Lens distortion*

```
r^2 = x_n^2 + y_n^2
radial = 1 + k1 r^2 + k2 r^4 + k3 r^6
x_d = x_n radial + 2 p1 x_n y_n + p2(r^2+2 x_n^2)
y_d = y_n radial + p1(r^2+2 y_n^2) + 2 p2 x_n y_n
```

Brown-Conrady model; fisheye optics require a suitable alternative.

### *Resolution and field of view*

```
GSD = H pixel_pitch/f = H sensor_width/(f N_width)
scene_width = 2 R tan(FOV_h/2)
R_min = target_width/(2 tan(FOV_h/2))
```

GSD assumes a plane normal to the viewing axis. For facade inspection, use normal stand-off range rather than altitude.

### *Stereo and uncertainty*

```
Z = f_px B/disparity_px
sigma_Z ~= Z^2 sigma_disparity/(f_px B)
```

Rectified calibrated stereo; uncertainty approximation assumes small disparity error.

### *Parallax and motion blur*

```
delta_u ~= f_x delta_X/Z
blur_px ~= f_px angular_rate exposure_time
exposure_time <= blur_limit/(f_px angular_rate)
```

Angular blur is a local approximation. Translational blur scales approximately as f_px V_perp exposure_time/Z; rolling shutter adds distortion.

### *Time to collision*

```
TTC = distance/closing_speed
TTC = image_scale/image_scale_rate
     = 1/(d ln(image_scale)/dt)
distance > v t_delay + v^2/(2 a_brake) + margin
```

Image expansion assumes a static object and suitable approach geometry. A normalized expansion rate, not arbitrary raw scale derivative, gives inverse TTC.

## 7. Optical flow and visual estimation

### *Normalized optical flow*

```
x_dot = -Vx/Z + x Vz/Z + x y wx - (1+x^2) wy + y wz
y_dot = -Vy/Z + y Vz/Z + (1+y^2) wx - x y wy - x wz
```

V and angular rates describe camera motion relative to a static scene in camera coordinates. Compensate rotation before inferring translational motion.

### *Reprojection and robust optimization*

```
r_i = z_i - project(T_CW P_W,i)
minimize sum_i robust_loss(r_i^T Sigma_i^(-1) r_i)
```

Moving objects, glass, repeated windows, darkness and low texture degrade estimation. Mask dynamic regions and monitor residuals and observability.

## 8. Visual-inertial navigation

### *IMU measurements*

```
a_m = R_BW(a_W-g_W) + b_a + n_a
omega_m = omega_true + b_g + n_g
b_a_dot = n_ba; b_g_dot = n_bg
```

Accelerometers measure specific force. Bias, scale factors, vibration and time offset affect integration.

### *Discrete propagation*

```
a_hat = R_k(a_m,k-b_a,k) + g_W
v_k+1 = v_k + a_hat dt
p_k+1 = p_k + v_k dt + 0.5 a_hat dt^2
q_k+1 = q_k (quaternion product) Exp_q((omega_m-b_g)dt)
```

Exp_q maps a rotation vector to [cos(angle/2), axis sin(angle/2)]. State that definition explicitly to avoid half-angle errors.

### *Uncertainty and measurement updates*

```
P_next = F P F^T + G Q G^T
S = H P H^T + R_measurement
K = P H^T S^(-1)
x_plus = x_minus + K(z-h(x_minus))
P_plus = (I-KH)P(I-KH)^T + K R_measurement K^T
innovation^T S^(-1) innovation < chi_square_threshold
```

Use error-state injection for quaternion states; the additive update is schematic. Joseph covariance form improves numerical robustness.

### *GNSS and vision*

```
z_GNSS = p_W + b_GNSS + n_GNSS
z_vision = project(T_CW P_W) + n_image
```

Urban non-line-of-sight GNSS errors may be biased and non-Gaussian. Monocular vision alone has scale ambiguity; VIO needs initialization and adequate excitation.

## 9. Fibre-tether architecture and mechanics

### *Architecture distinction*

```
anchored: cable between fixed/moving anchor and vehicle
payout: onboard spool releases fibre along the flight path
```

A fibre-only payout link does not inherently supply power or persistent endurance. An anchored hybrid tether may combine fibre, power conductors and a strength member. Do not apply an anchored catenary model blindly to

deposited fibre.

### *Taut-link force and constraint*

```
r = p_U - p_A; distance = |r|; r_hat = r/|r|
F_tether = -Tension r_hat; Tension >= 0
|r| <= L_deployed
r dot v_relative = 0
r dot a_relative + |v_relative|^2 = 0
```

Last two equations assume a taut, inextensible cable of constant length. Include payout-rate terms for a winch with changing deployed length.

### *Elastic approximation*

```
extension = max(0, distance-L0)
k_t = E_effective A_cs/L0
Tension = max(0, k_t extension + c_t extension_rate)
```

Effective axial stiffness belongs to the cable assembly, not automatically the glass fibre. A spring-damper model does not capture distributed sag.

### *Catenary and small sag*

```
w = mu g; a = H_tension/w
z(x) = a cosh((x-x0)/a) + z0
L = 2 a sinh(X/(2 a)) [equal-height endpoints]
sag = a [cosh(X/(2 a))-1]
T(x) = H_tension cosh((x-x0)/a)
sag ~= w X^2/(8 H_tension)
```

Ideal static uniform cable under gravity. Unequal endpoint heights require boundary-condition solving; wind changes the load direction and shape.

### *Weight, drag and endpoint loading*

```
m_tether = mu L; W_tether = mu L g
dF_drag = 0.5 rho C_D,t d_t V_perp^2 ds
F_drag,total ~= 0.5 rho C_D,t d_t L V_perp^2
```

For a straight near-vertical freely hanging cable, upper-end weight load approaches W_tether. For two supported endpoints, solve load sharing; do not assign all weight or drag to the aircraft by default.

### *Strength and vibration*

```
T_peak <= min(T_allow, T_break/SF)
f_mode,n = n/(2 L) sqrt(Tension/mu)
c_wave = sqrt(Tension/mu)
t_roundtrip = 2 L/c_wave
```

Vibration equations assume uniform tension and ideal fixed ends. Evaluate peak dynamic loads rather than adding incompatible maxima. Snags require separate transient and breakaway analysis.

## 10. Fibre optics and power tether

### *Optical budget*

```
P_out,dBm = P_in,dBm - alpha_dB_per_km L_km
            - L_connector - L_splice - L_bend
M_optical = P_out - sensitivity - M_system
```

```
t_fibre = n_group L/c ~= 5 microseconds/km
```

Check transceiver compatibility, bandwidth, optical budget, bend radius and spool losses. Fibre reduces RF dependence but not mechanical failure risk.

### *Hybrid electrical tether*

```
R_loop = 2 resistivity L/A_conductor
V_drop = I R_loop; P_loss = I^2 R_loop
P_delivered = V_source I - P_loss
```

Applies only when conductors are present. Include conversion efficiency, thermal limits, insulation, fault protection and conductor mass.

## 11. Clearance, control and feasibility

### *Containment margins*

```
d_required = d_stop + d_position_error + d_wind_drift
             + d_map_error + d_safety
d_drift = v_lateral,0 t_response + 0.5 a_w t_response^2
```

Use uncertainty-aware margins and obstacle geometry, not a single scalar geofence distance. Include the swept tether volume and possible snag points.

### *Position and attitude commands*

```
e_p = p_ref-p; e_v = v_ref-v
a_cmd = Kp e_p + Kd e_v + Ki integral(e_p dt) + a_ff
F_cmd = m(a_cmd-g_W) - F_disturbance_estimate
T_cmd = |F_cmd|; b3_desired = F_cmd/|F_cmd|
e_R = 0.5 vee(R_d^T R - R^T R_d)
e_omega = omega - R^T R_d omega_d
tau = -K_R e_R - K_omega e_omega + omega cross(I omega)
      + feedforward_terms
```

Define the actual attitude controller and actuator allocation before implementation. Apply rotor, tilt, rate, power and thermal saturation plus anti-windup.

### *Feasibility gates*

```
T_required <= gamma_T T_max, where 0 < gamma_T < 1
E_mission + E_return + E_contingency <= E_usable
M_RF >= required_margin; M_optical >= required_margin
T_peak <= rated_working_load
minimum_clearance >= uncertainty_and_response_margin
```

For vectors, required propulsive force is m(a_cmd-g_W)-F_aero-F_tether. CV acceptance must test feature geometry, residuals, blur, uncertainty and recovery behavior, not just feature count.

## 12. Operational workflow

1. Define frames, vehicle configuration, propulsion maps, payload and electrical limits.

2. Build obstacle, facade, corridor, rooftop and anchor geometry with uncertainty.

3. Measure or bound local wind and gusts; distinguish weather-station wind from facade flow.

4. Evaluate hover, maneuver, braking and disturbance forces with power and tilt limits.

5. Integrate mission energy and reserve requirements.

6. Survey radio coverage, interference, antenna orientation and link-loss recovery.

7. Validate camera calibration, timing, blur, stereo range and VIO initialization.

8. Identify anchored versus payout fibre architecture; model cable loads, routing, spool and bend limits.

9. Check vehicle and cable swept volumes, contingency landing and snag recovery.

10. Validate with controlled tests, hardware logs, simulation and fault-injection exercises.

## 13. Illustrative parameter schema

```
scenario:
  coordinate_frame: ENU
  mission_mode: CV-navigation
  altitude_AGL_m: 45
  target_speed_m_s: 6
  wind_ENU_m_s: [4, -2, 0]
vehicle:
  mass_kg: 4.2
  rotor_count: 4
  propeller_diameter_m: 0.43
  maximum_total_thrust_N: 130
  battery_voltage_V: 44.4
  battery_capacity_Ah: 16
  usable_depth_of_discharge: 0.8
radio:
  frequency_MHz: 2400
  transmit_power_dBm: 20
  receiver_sensitivity_dBm: -98
  fade_margin_dB: 15
  implementation_margin_dB: 5
  end_to_end_latency_s: 0.12
vision:
  resolution_px: [3840, 2160]
  focal_length_px: [2100, 2100]
  principal_point_px: [1920, 1080]
  frame_rate_Hz: 30
  exposure_time_s: 0.003
  stereo_baseline_m: 0.12
tether:
  architecture: anchored-hybrid
  enabled: false
  deployed_length_m: 80
  diameter_m: 0.006
  linear_mass_density_kg_m: 0.045
  effective_axial_stiffness_N: null
  breaking_strength_N: 3000
  allowable_tension_N: 750
  safety_factor: 4
  fibre_attenuation_dB_per_km: 0.35
validation:
  parameter_status: illustrative-not-certified
  camera_FOV: derive-from-calibrated-intrinsics
  flight_limits: obtain-from-manufacturer-and-testing
```

## 14. Safety and interpretation

This skill supports engineering analysis, simulation and test planning. It does not replace manufacturer limits, airworthiness assessment, local aviation rules, site permissions or a formal safety case. Calibrate thrust, drag, electrical losses, radio fade, camera timing and cable properties on the actual system.

Editorial clarifications versus the original draft: removed unsupported universal thrust and ground-effect thresholds; corrected normalized time-to-collision interpretation; separated cable endpoint loads from whole-system loads; and distinguished fibre payout from powered anchored tethers. No performance values in the illustrative schema are operational approvals.

## 15. Source context retained from the original lookup

NASA, Estimation, Navigation and Control of Multi-Rotor Drones in an Urban Environment
https://ntrs.nasa.gov/api/citations/20180008712/downloads/20180008712.pdf

DTU, Vision-based Positioning for UAVs
https://backend.orbit.dtu.dk/ws/files/245930144/Thesis_Xiao_Hu.pdf

Modelling and Simulation of a Tethered UAS
https://staff.polito.it/giorgio.guglieri/Flight_Mechanics_Research_Team/Papers_files/20_0198.pdf

Tethered Drones: A Comprehensive Review
https://www.research.unipd.it/retrieve/94784c42-a6f5-4c8e-9ef3-0509f9aff1c8/drones-09-00425_REVIEW_TETHERED_DRONES.pdf

Survey of Air-to-Ground Propagation Channel Modeling
https://people.computing.clemson.edu/~jmarty/projects/lowLatencyNetworking/papers/LossandPropModels/ForUAVSystems/SurveyOfAirToGndChannelModelingForDrones.pdf